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Left Atrial Appendage Occluder Devices

Abstract

A left atrial appendage (“LAA”) occluder may include a proximal disk configured to cover an ostium of the LAA, a distal lobe configured to be received within a cavity of the LAA, and a connecting member connecting the proximal disk to the distal lobe. The occluder may include first and second groups of stabilizing members having first and second engagement features, respectively, configured to engage tissue surrounding the LAA when the occluder is implanted. In an unbiased condition of the occluder, the distal lobe may have an axial length extending between a proximal face and a distal face of the distal lobe. The second engagement features may be positioned at least 33% of the axial length beyond the proximal face of the distal lobe, and the first engagement features may be positioned distal to the second engagement features.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority to the filing dates of (i) U.S. Provisional Patent Application No. 63/694,087, filed Sep. 12, 2024, (ii) U.S. Provisional Patent Application No. 63/645,351, filed May 10, 2024, and (iii) U.S. Provisional Patent Application No. 63/562,342, filed Mar. 7, 2024, the disclosures of all of which are hereby incorporated by reference herein.

FIELD OF THE DISCLOSURE

[0002] The present disclosure relates generally to medical devices that are used in the human body. In particular, the present disclosure is directed to an occlusion device having a configuration that allows for more consistent and stable anchoring of the occlusion device within a tissue cavity. More specifically, the present disclosure is directed to an occlusion device with one or more rows of hooks or stabilizing wires that increase the resilience of the device to embolization or motion in the implanted condition.

BACKGROUND

[0003] An occluder is a medical device used to treat (e.g., occlude) tissue at a target site within the human body, such as an abnormality, a vessel, an organ, an opening, a chamber, a channel, a hole, a cavity, a lumen, or the like. For example, an occluder may be used for Left Atrial Appendage (“LAA”) closures. An LAA is a normal anatomical structure which is a sac or pouch in the muscle wall of the left atrium. When a patient experiences atrial fibrillation (“AFib”), a blood clot may be formed within the LAA which may become dislodged and enter into the blood stream. By occluding the LAA, the release of blood clots from the LAA may be significantly reduced, if not eliminated. Various techniques have been developed to occlude the LAA. For instance, balloon-like devices have been developed that are configured to be implanted completely within the cavity of the LAA, while surgical techniques have also been developed where the cavity of the LAA is inverted and surgically closed.

[0004] Despite these techniques, it would be advantageous to provide an improved occlusion device that offers a reduced risk of adverse events such as embolization.

SUMMARY OF THE DISCLOSURE

[0005] According to one aspect of the disclosure, a medical device is for occluding a left atrial appendage (“LAA”). The medical device may include a proximal disk configured to cover an ostium of the LAA in an implanted condition of the medical device, a distal lobe configured to be received within a cavity of the LAA in the implanted condition of the medical device, and a connecting member connecting the proximal disk to the distal lobe. The medical device may include a first group of stabilizing wires, each stabilizing wire in the first group having a first hooked end configured to engage tissue surrounding the LAA in the implanted condition of the medical device. The medical device may include a second group of stabilizing wires, each stabilizing wire in the second group having a second hooked end configured to engage tissue surrounding the LAA in the implanted condition of the medical device. In an unbiased condition of the medical device, the distal lobe may have an axial length extending between a proximal face and a distal face of the distal lobe. The second hooked ends of the stabilizing wires in the second group may be positioned at least one-third of the length beyond the proximal face of the distal lobe. The

first hooked ends of the stabilizing wires in the first group may be positioned distal to the second hooked ends of the stabilizing wires in the second group. The second hooked ends of the stabilizing wires in the second group may be positioned between about 33% and about 55% of the axial length beyond the proximal face of the distal lobe. The axial length may be about 7.5 mm, and the second hooked ends of the stabilizing wires in the second group may be positioned between about 3.0 mm and about 4.0 mm beyond the proximal face of the distal lobe. The axial length may be about 10 mm, and the second hooked ends of the stabilizing wires in the second group may be positioned between about 3.75 mm and about 5.25 mm beyond the proximal face of the distal lobe. Each stabilizing wire in the first group may be positioned between a pair of stabilizing wires in the second group. Each stabilizing wire in the first group may be substantially identical in structure to each stabilizing wire in the second group. Each stabilizing wire in the first group may have an axial length that is equal to or longer than an axial length of each stabilizing wire in the second group. A total number of the stabilizing wires in the first group may be equal to a total number of stabilizing wires in the second group. A spacing between each adjacent pair of stabilizing wires may be substantially equal. A radial spacing between each adjacent pair of stabilizing wires may be between about 20 degrees and about 65 degrees.

[0006] According to another aspect of the disclosure, a medical device is for occluding a left atrial appendage (“LAA”). The medical device may include a proximal disk configured to cover an ostium of the LAA in an implanted condition of the medical device, a distal lobe configured to be received within a cavity of the LAA in the implanted condition of the medical device, a connecting member connecting the proximal disk to the distal lobe, and a plurality of stabilizing wires. The plurality of stabilizing wires may include a first group of hooked ends configured to engage tissue surrounding the LAA in the implanted condition of the medical device and a second group of hooked ends configured to engage tissue surrounding the LAA in the implanted condition of the medical device. In an unbiased condition of the medical device, the distal lobe may have an axial length extending between a proximal face and a distal face of the distal lobe, and the second group of hooked ends of the plurality of stabilizing wires may be positioned at least 33% of the axial length beyond the proximal face of the distal lobe. The first group of hooked ends of the plurality of stabilizing wires may be positioned distal to the second group of hooked ends of the plurality of stabilizing wires. The plurality of stabilizing wires may include a first group of stabilizing wires, and a second group of stabilizing wires. The first group of hooked ends may be positioned on the first group of stabilizing wires, the second group of hooked ends may be positioned on the second group of stabilizing wires. Each of the plurality of stabilizing wires may include at least one hooked end of the first group of hooked ends, and at least one hooked end of the second group of hooked ends. Each of the plurality of stabilizing wires may include one pair of hooked ends of the first group of hooked ends, and one pair of hooked ends of the second group of hooked ends. Each of the plurality of stabilizing wires may be isolated or separate from the other stabilizing wires of the plurality of stabilizing wires. Each group of stabilizing wires may comprise more than one stabilizing wire. The plurality of stabilizing wires may be coupled to, or connected to, the distal lobe. Each of the plurality of stabilizing wires may include a backing portion. Each backing portion may be U-shaped. Each backing portion may comprise one or more eyelets. A pair of eyelets may be provided for the respective pair of hooked ends of each backing portion. The eyelets of the plurality of stabilizing wires may be coupled to, or connected to, the distal lobe. Each of the plurality of stabilizing wires may include a backing portion, the backing portion including two legs extending therefrom, each of the two legs extending to a terminal end, the terminal end of each leg forming one of the pair of hooked ends of the first group of hooked ends, each of the pair of hooked ends of the second group of hooked ends extending from a respective one of the two legs. Each of the plurality of stabilizing wires may include a single hooked end of the first group of hooked ends, and a single hooked end of the second group of hooked ends. Each of the plurality of stabilizing wires may include a backing portion, the backing portion including two legs extending

therefrom, a first one of the two legs terminating in the single hooked end of the first group of hooked ends, and a second one of the two legs terminating in the single hooked end of the second group of hooked ends, the first one of the two legs having a length that is longer than a length of the second one of the two legs. Each of the plurality of stabilizing wires may include a pair of hooked ends of the first group of hooked ends, and a single hooked end of the second group of hooked ends. Each of the plurality of stabilizing wires may include a backing portion, the backing portion including two legs extending therefrom, each of the two legs extending to a terminal end, the terminal end of each leg forming one of the pair of hooked ends of the first group of hooked ends, the single hooked end of the second group of hooked ends being formed integrally with the backing portion. Each of the plurality of stabilizing wires may include a backing portion, the backing portion including two legs extending therefrom, each of the two legs extending to a terminal end, the terminal end of each leg forming one of the pair of hooked ends of the first group of hooked ends, the single hooked end of the second group of hooked ends being coupled, via a connector, to the backing portion.

[0007] According to another aspect of the disclosure, a medical device is for occluding a left atrial appendage (“LAA”). The medical device may include a proximal disk configured to cover an ostium of the LAA in an implanted condition of the medical device, a distal lobe configured to be received within a cavity of the LAA in the implanted condition of the medical device, and a connecting member connecting the proximal disk to the distal lobe. The medical device may also include a first group of stabilizing wires, each stabilizing wire in the first group having a first hooked end configured to engage tissue surrounding the LAA in the implanted condition of the medical device. The medical device may also include a second group of stabilizing wires, each stabilizing wire in the second group having a second hooked end configured to engage tissue surrounding the LAA in the implanted condition of the medical device. The medical device may also include a third group of stabilizing wires, each stabilizing wire in the third group having a third hooked end configured to engage tissue surrounding the LAA in the implanted condition of the medical device. In an unbiased condition of the medical device, the distal lobe may have an axial length extending between a proximal face and a distal face of the distal lobe. The first hooked ends may be positioned at a first distance along the axial length. The second hooked ends may be positioned at a second distance along the axial length. The third hooked ends may be positioned at a third distance along the axial length. The first distance may be different than the second distance. The second distance may be different than the third distance. The third distance may be different than the first distance. The axial length may be about 10 mm. The first distance may be between about 8 mm and about 10 mm, such that the first hooked ends are positioned between about 0 mm and about 2 mm proximally of the distal face of the distal lobe. The second distance may be between about 3.3 mm and about 4.3 mm. The third distance may be between about 4.4 mm and about 5.4 mm. The medical device may include a total number of stabilizing wires in the first group that is at least twice as many as a total number of stabilizing wires in the second group and that is at least twice as many as a total number of stabilizing wires in the third group. A single stabilizing wire in either the second group or the third group may be positioned between each circumferentially adjacent pair of stabilizing wires in the first group. Each stabilizing wire may be separate from the other stabilizing wires. Each group of stabilizing wires may comprise more than one stabilizing wire. Each stabilizing wire may be connected to the distal lobe.

[0008] According to a further aspect of the disclosure there is provided a medical device for occluding a left atrial appendage (“LAA”), the medical device comprising: a proximal disk configured to cover an ostium of the LAA in an implanted condition of the medical device; a distal lobe configured to be received within a cavity of the LAA in the implanted condition of the medical device; a connecting member connecting the proximal disk to the distal lobe; and a plurality of stabilizing wires, the plurality of stabilizing wires including a first group of hooked ends configured to engage tissue surrounding the LAA in the implanted condition of the medical device

and a second group of hooked ends configured to engage tissue surrounding the LAA in the implanted condition of the medical device, wherein in an unbiased condition of the medical device, the distal lobe has an axial length extending between a proximal face and a distal face of the distal lobe, the first group of hooked ends of the plurality of stabilizing wires being positioned distal to the second group of hooked ends of the plurality of stabilizing wires. The second group of hooked ends of the plurality of stabilizing wires may be positioned at least 33% of the axial length beyond the proximal face of the distal lobe. The plurality of stabilizing wires may comprise a first group of stabilizing wires and a second group of stabilizing wires. Each stabilizing wire in the first group of stabilizing wires may have a hooked end to provide the first group of hooked ends. Each stabilizing wire in the second group of stabilizing wires may have a hooked end to provide the second group of hooked ends. The plurality of stabilizing wires may comprise a third group of stabilizing wires. Each stabilizing wire in the third group may have a third hooked end configured to engage tissue surrounding the LAA in the implanted condition of the medical device. The first hooked ends may be positioned at a first distance along the axial length, the second hooked ends may be positioned at a second distance along the axial length, and the third hooked ends may be positioned at a third distance along the axial length. The first distance may be different than the second distance, the second distance may be different than the third distance, and the third distance may be different than the first distance.

[0009] According to a further aspect of the disclosure there is provided a medical device comprising a lobe and a plurality of stabilizing wires that are connected to the lobe, the plurality of stabilizing wires including a first group of hooked ends configured to engage tissue surrounding the lobe in an implanted condition of the medical device and a second group of hooked ends configured to engage tissue surrounding the lobe in the implanted condition of the medical device, wherein, in an unbiased condition of the medical device, the lobe has an axial length extending between a proximal face and a distal face of the distal lobe, and wherein the first group of hooked ends of the plurality of stabilizing wires are positioned distal to the second group of hooked ends of the plurality of stabilizing wires.

[0010] According to one aspect of the disclosure, a medical device for occluding a left atrial appendage (“LAA”) includes a proximal disk configured to cover an ostium of the LAA in an implanted condition of the medical device, a distal lobe configured to be received within a cavity of the LAA in the implanted condition of the medical device, and a connecting member connecting the proximal disk to the distal lobe. The medical device includes a first group of stabilizing members, each stabilizing member in the first group having a first engagement feature configured to engage tissue surrounding the LAA in the implanted condition of the medical device. The medical device includes a second group of stabilizing members, each stabilizing member in the second group having a second engagement feature configured to engage tissue surrounding the LAA in the implanted condition of the medical device. In an unbiased condition of the medical device, the distal lobe has an axial length extending between a proximal face and a distal face of the distal lobe, the second engagement features of the stabilizing members in the second group being positioned at least 33% of the axial length beyond the proximal face of the distal lobe, the first engagement features of the stabilizing members in the first group being positioned distal to the second engagement features of the stabilizing members in the second group. The second engagement features of the stabilizing members in the second group may be positioned between about 33% and about 55% of the axial length beyond the proximal face of the distal lobe. The axial length may be about 7.5 mm, and the second engagement features of the stabilizing members in the second group may be positioned between about 3.0 mm and about 4.0 mm beyond the proximal face of the distal lobe. The axial length may be about 10 mm, and the second engagement features of the stabilizing members in the second group may be positioned between about 3.75 mm and about 5.25 mm beyond the proximal face of the distal lobe. Each stabilizing member in the first group may be positioned between a pair of stabilizing members in the second group. Each

stabilizing member in the first group may be substantially identical in structure to each stabilizing member in the second group. Each stabilizing member in the first group may have an axial length that is equal to or longer than an axial length of each stabilizing member in the second group. A total number of the stabilizing members in the first group may be equal to a total number of stabilizing members in the second group. A spacing between each adjacent pair of stabilizing members may be substantially equal. A radial spacing between each adjacent pair of stabilizing members may be between about 20 degrees and about 65 degrees.

[0011] According to another aspect of the disclosure, a medical device for occluding a left atrial appendage (“LAA”) includes a proximal disk configured to cover an ostium of the LAA in an implanted condition of the medical device, a distal lobe configured to be received within a cavity of the LAA in the implanted condition of the medical device, and a connecting member connecting the proximal disk to the distal lobe. The medical device includes a plurality of stabilizing members, the plurality of stabilizing members including a first group of engagement features configured to engage tissue surrounding the LAA in the implanted condition of the medical device and a second group of engagement features configured to engage tissue surrounding the LAA in the implanted condition of the medical device. In an unbiased condition of the medical device, the distal lobe has an axial length extending between a proximal face and a distal face of the distal lobe, the second group of engagement features of the plurality of stabilizing members being positioned at least 33% of the axial length beyond the proximal face of the distal lobe, the first group of engagement features of the plurality of stabilizing members being positioned distal to the second group of engagement features of the plurality of stabilizing members. The plurality of stabilizing members may include a first group of stabilizing members, and a second group of stabilizing members, the first group of engagement features being positioned on the first group of stabilizing members, the second group of engagement features being positioned on the second group of stabilizing members. Each of the plurality of stabilizing members may include at least one engagement feature of the first group of engagement features, and at least one engagement feature of the second group of engagement features. Each of the plurality of stabilizing members may include one pair of engagement features of the first group of engagement features, and one pair of engagement features of the second group of engagement features. Each of the plurality of stabilizing members may be isolated or separate from the other stabilizing members of the plurality of stabilizing members. Each group of stabilizing members may comprise more than one stabilizing member. The plurality of stabilizing members may be coupled to, or connected to, the distal lobe. Each of the plurality of stabilizing members may include a backing portion. Each backing portion may be U-shaped. Each backing portion may comprise one or more eyelets. A pair of eyelets may be provided for the respective pair of engagement features of each backing portion. The eyelets of the plurality of stabilizing members may be coupled to, or connected to, the distal lobe. Each of the plurality of stabilizing members may include a backing portion, the backing portion including two legs extending therefrom, each of the two legs extending to a terminal end, the terminal end of each leg forming one of the pair of engagement features of the first group of engagement features, each of the pair of engagements features of the second group of engagement features extending from a respective one of the two legs. Each of the plurality of stabilizing members may include a single engagement feature of the first group of engagement features, and a single engagement feature of the second group of engagement features. Each of the plurality of stabilizing members may include a backing portion, the backing portion including two legs extending therefrom, a first one of the two legs terminating in the single engagement feature of the first group of engagement features, and a second one of the two legs terminating in the single engagement feature of the second group of engagement features, the first one of the two legs having a length that is longer than a length of the second one of the two legs. Each of the plurality of stabilizing members may include a pair of engagement features of the first group of engagement features, and a single engagement feature of the second group of engagement features. Each of the plurality of stabilizing members may include

a backing portion, the backing portion including two legs extending therefrom, each of the two legs extending to a terminal end, the terminal end of each leg forming one of the pair of engagement features of the first group of engagement features, the single engagement feature of the second group of engagement features being formed integrally with the backing portion. Each of the plurality of stabilizing members may include a backing portion, the backing portion including two legs extending therefrom, each of the two legs extending to a terminal end, the terminal end of each leg forming one of the pair of engagement features of the first group of engagement features, the single engagement feature of the second group of engagement features being coupled, via a connector, to the backing portion.

[0012] According to a further aspect of the disclosure, a medical device for occluding a left atrial appendage (“LAA”) includes a proximal disk configured to cover an ostium of the LAA in an implanted condition of the medical device, a distal lobe configured to be received within a cavity of the LAA in the implanted condition of the medical device, and a connecting member connecting the proximal disk to the distal lobe. The medical device includes a first group of stabilizing members, each stabilizing member in the first group having a first engagement feature configured to engage tissue surrounding the LAA in the implanted condition of the medical device. The medical device includes a second group of stabilizing members, each stabilizing member in the second group having a second engagement feature configured to engage tissue surrounding the LAA in the implanted condition of the medical device. The medical device includes a third group of stabilizing members, each stabilizing member in the third group having a third engagement feature configured to engage tissue surrounding the LAA in the implanted condition of the medical device. In an unbiased condition of the medical device, the distal lobe has an axial length extending between a proximal face and a distal face of the distal lobe, the first engagement features are positioned at a first distance along the axial length, the second engagement features are positioned at a second distance along the axial length, and the third engagement features are positioned at a third distance along the axial length, the first distance being different than the second distance, the second distance being different than the third distance, the third distance being different than the first distance. The axial length may be about 10 mm. The first distance may be between about 8 mm and about 10 mm, such that the first engagement features are positioned between about 0 mm and about 2 mm proximally of the distal face of the distal lobe. The second distance may be between about 3.3 mm and about 4.3 mm. The third distance may be between about 4.4 mm and about 5.4 mm. The medical device may include a total number of stabilizing members in the first group that is at least twice as many as a total number of stabilizing members in the second group and that is at least twice as many as a total number of stabilizing members in the third group. A single stabilizing member in either the second group or the third group may be positioned between each circumferentially adjacent pair of stabilizing members in the first group. Each stabilizing member may be separate from the other stabilizing members, and each group of stabilizing members may comprise more than one stabilizing member, and each stabilizing member may be connected to the distal lobe. Each stabilizing member in the first, second, and third groups may be formed of a wire. Each stabilizing member in the first, second, and third groups may be cut from a sheet of material. Each first engagement feature, each second engagement feature, and each third engagement feature may be a hook.

[0013] According to a further aspect of the disclosure there is provided a medical device for occluding a left atrial appendage (“LAA”), the medical device comprising: a proximal disk configured to cover an ostium of the LAA in an implanted condition of the medical device; a distal lobe configured to be received within a cavity of the LAA in the implanted condition of the medical device; a connecting member connecting the proximal disk to the distal lobe; and a plurality of stabilizing members, the plurality of stabilizing members including a first group of engagement features configured to engage tissue surrounding the LAA in the implanted condition of the medical device and a second group of engagement features configured to engage tissue surrounding the

LAA in the implanted condition of the medical device, wherein in an unbiased condition of the medical device, the distal lobe has an axial length extending between a proximal face and a distal face of the distal lobe, the first group of engagement features of the plurality of stabilizing members being positioned distal to the second group of engagement features of the plurality of stabilizing members. The second group of engagement features of the plurality of stabilizing members may be positioned at least 33% of the axial length beyond the proximal face of the distal lobe. The plurality of stabilizing members may comprise a first group of stabilizing members and a second group of stabilizing members. Each stabilizing member in the first group of stabilizing members may have an engagement feature to provide the first group of engagement features. Each stabilizing member in the second group of stabilizing members may have an engagement feature to provide the second group of engagement features. The plurality of stabilizing members may comprise a third group of stabilizing members. Each stabilizing member in the third group may have a third engagement feature configured to engage tissue surrounding the LAA in the implanted condition of the medical device. The first engagement features may be positioned at a first distance along the axial length, the second engagement features may be positioned at a second distance along the axial length, and the third engagement features may be positioned at a third distance along the axial length. The first distance may be different than the second distance, the second distance may be different than the third distance, and the third distance may be different than the first distance.

[0014] According to a further aspect of the disclosure there is provided a medical device comprising a lobe and a plurality of stabilizing members that are connected to the lobe, the plurality of stabilizing members including a first group of engagement features configured to engage tissue surrounding the lobe in an implanted condition of the medical device and a second group of engagement features configured to engage tissue surrounding the lobe in the implanted condition of the medical device, wherein, in an unbiased condition of the medical device, the lobe has an axial length extending between a proximal face and a distal face of the distal lobe, and wherein the first group of engagement features of the plurality of stabilizing members are positioned distal to the second group of engagement features of the plurality of stabilizing members.

[0015] It will be appreciated that, in general, a feature described in relation to one embodiment may be provided in combination with another embodiment.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0016] FIG. 1 illustrates a known medical device.

[0017] FIGS. 2A-2C are a schematic diagram of the known medical device shown in FIG. 1 under radial compression.

[0018] FIG. 3 is a schematic diagram of a delivery system in accordance with the present disclosure.

[0019] FIG. 4 is a highly schematic view of the medical device of FIG. 1 implanted within an exemplary LAA.

[0020] FIGS. 5-6 show an example of an occluder according to an aspect of the disclosure, the occluder being in an expanded condition.

[0021] FIGS. 7A-B are exemplary embodiments of a stabilizing member with at least one eyelet formed by wire or by laser cut design in accordance with the present disclosure.

[0022] FIG. 8 is an enlarged view of a stabilizing member connected to an occluder.

[0023] FIG. 9A is another view of an occluder similar to that shown in FIGS. 5A-B, with FIG. 9B showing a pattern for the smaller stabilizing member before being shaped or contoured.

[0024] FIG. 10A is another example of an occluder with two rows of engagement features of stabilizing members at different axial positions, with FIGS. 10B-10D showing examples of patterns for the stabilizing member before being shaped or contoured.

[0025] FIG. 11 is another example of an occluder with two rows of engagement features of stabilizing members at different axial positions.

[0026] FIG. 12A is another example of an occluder with two rows of engagement features of stabilizing members at different axial positions, with FIGS. 12B-12C showing examples of the stabilizing members.

[0027] FIG. 13 is another example of an occluder with two rows of engagement features of stabilizing members at different axial positions, including two rows of smaller stabilizing members similar to the smaller stabilizing members shown in FIGS. 5-6.

[0028] FIGS. 14-15 are top perspective and bottoms views, respectively, of an occluder according to another aspect of the disclosure.

[0029] FIG. 16 is a side view of the occluder of FIGS. 14-15.

[0030] FIG. 17 is a simplified side view of the occluder of FIGS. 14-16.

DETAILED DESCRIPTION OF THE DISCLOSURE

[0031] The present disclosure relates generally to medical devices that are used in the human body. Specifically, the present disclosure provides medical devices including occlusion devices having stabilizing members for enhancing engagement of the occluder with the tissue in which it is implanted. The disclosed embodiments may lead to more consistent and improved patient outcomes. It is contemplated, however, that the described features and methods of the present disclosure as described herein may be incorporated into any number of systems as would be appreciated by one of ordinary skill in the art based on the disclosure herein.

[0032] Although the exemplary embodiment of the medical device is described as treating a target site including a LAA, it is understood that the use of the term “target site” is not meant to be limiting, as the medical device may be configured to treat any target site, such as an abnormality, a vessel, an organ, an opening, a chamber, a channel, a hole, a cavity, or the like, located anywhere in the body. The term “vascular abnormality,” as used herein is not meant to be limiting, as the medical device may be configured to bridge or otherwise support a variety of vascular abnormalities. For example, the vascular abnormality could be any abnormality that affects the shape of the native lumen, such as an atrial septal defect, a lesion, a vessel dissection, or a tumor. Embodiments of the medical device may be useful, for example, for occluding a patent foramen ovalis (“PFO”), atrial septal defect (“ASD”), ventricular septal defect (“VSD”), or patent ductus arteriosus (“PDA”), as noted above. Furthermore, the term “lumen” is also not meant to be limiting, as the vascular abnormality may reside in a variety of locations within the vasculature, such as a vessel, an artery, a vein, a passageway, an organ, a cavity, or the like. As used herein, the term “proximal” refers to a part of the medical device or the delivery device that is closest to the operator, and the term “distal” refers to a part of the medical device or the delivery device that is farther from the operator at any given time as the medical device is being delivered through the delivery device. In addition, the terms “deployed” and “implanted” may be used interchangeably herein.

[0033] Some embodiments of the present disclosure provide an improved percutaneous catheter directed intravascular occlusion device for use in the vasculature in patients' bodies, such as blood vessels, channels, lumens, a hole through tissue, cavities, and the like, such as a LAA. Other physiologic conditions in the body occur where it is also desirous to occlude a vessel or other passageway to prevent blood flow into or therethrough. These device embodiments may be used anywhere in the vasculature where the anatomical conditions are appropriate for the design.

[0034] The medical device may include one or more layers of occlusive material, wherein each layer may be comprised of any material that is configured to substantially preclude or occlude the flow of blood so as to facilitate thrombosis. As used herein, “substantially preclude or occlude

flow” shall mean, functionally, that blood flow may occur for a short time, but that the body's clotting mechanism or protein or other body deposits on the occlusive material results in occlusion or flow stoppage after this initial time period.

[0035] Some embodiments of the present disclosure may be formed by a plurality of wire strands having a predetermined relative orientation with respect to one another. However, it is understood that according to additional embodiments of the present disclosure, that the medical device could be etched or laser cut from a tube, or the device could comprise an occlusion material coupled to a scaffolding structure or a plurality of slices of a tubular member coupled together.

[0036] The present disclosure now will be described more fully hereinafter with reference to the accompanying drawings, in which some, but not all embodiments of the disclosure are shown. Indeed, this disclosure may be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will satisfy applicable legal requirements. Like numbers refer to like elements throughout.

[0037] In at least some conventional or known medical devices used for the occlusion of abnormalities, such as a medical device **50** shown in FIG. **1**, medical device **50** includes a proximal end **52** and a distal end **54**, with a disk **56** at proximal end **52** and a lobe **58** at distal end **54**. The lobe **58** has a proximal edge **60** (also referred to as a proximal face), a distal edge **62** (also referred to as a distal face), and a middle or central portion **64** that define a cavity **66**. The medical device **50** also includes stabilizing members (which may be formed using wire(s)) **68** secured to a radially outer or circumferential surface of middle portion **64**. The stabilizing members **68** terminate in an engagement feature (which may be a hook or a hooked end of a wire forming the stabilizing member) **70** at free ends thereof, and thereby facilitate retention of the medical device **50** at a target site and preventing the medical device **50** from becoming dislodged from the target site after deployment.

[0038] In this known medical device **50**, proximal edge **60** and distal edge **62** adjoin middle portion **64** at a first relatively blunt or sharp (e.g., non-rounded) transition **72** and a second blunt transition **74**, respectively. First blunt transition **72** connects proximal edge **60** to middle portion **64** by an approximately 90 degree angle. Likewise, second blunt transition **74** connects distal edge **62** to middle portion **64** by an approximately 90 degree angle. First blunt transition **72** and second blunt transition **74** partially define a generally rectangular cross section to lobe **58**, leading to relatively blunt circumferential edges of the device and relatively high radial force applied to the surrounding tissue.

[0039] Turning now to FIGS. **2A-2C**, medical device **50** before and after undergoing radial compression is depicted. Before radial compression (FIG. **2A**) is applied to lobe **58**, the outer surface of middle portion **64** is linear or extends generally perpendicular to proximal and distal faces **60**, **62**. Each engagement feature (e.g. hook) **70** of a corresponding stabilizing member (e.g. stabilizing wire) **68** is at a first angle **76** with respect to a generally longitudinal direction **77**. When radial compression is applied to lobe **58** (FIGS. **2B** and **2C**), proximal and distal faces **60**, **62** flex and bow outwardly (e.g., axially outward), and middle portion **64** of lobe **58** flexes and bows inwardly, in response to the applied force. The approximately 90 degree angle of first blunt transition **72** and second blunt transition **74** force the outer surface of middle portion **64** to transition from linear to concave when proximal and distal faces **60**, **62** bow outwardly. The concave shape adopted by lobe **58** also shifts the position of stabilizing members **68**, such that stabilizing members **68** at least partially contract and engagement features **70** transition from first angle **76** to a second, greater angle **78**. At second angle **78**, engagement features **70** are oriented more directly towards the adjacent tissue, than when engagement features **70** are at first angle **76**. This shift in orientation of stabilizing members **68**, and therefore engagement members **70**, can lead to an increase in interactions between engagement members **70** and the adjacent tissue at the target site within the patient's body. The increased interaction with tissue can lead to adverse side effects

such as late pericardial effusion.

[0040] The medical devices of the present disclosure may include a rounded lobe, for example as described in greater detail in U.S. Patent Application Publication No. 2022/0008050, the disclosure of which is hereby incorporated by reference herein. The use of a more rounded lobe may lead to a more uniform radial compression, reduction in radial force applied to surrounding tissue, and reduction in variability of the hook angle of the stabilizing wires, minimizing potential disadvantages of known medical devices.

[0041] Turning now to FIG. 3, a schematic diagram of a delivery system **100** is shown. Delivery system **100** includes a delivery device **102** including a catheter **104** and a coupling member **106** configured to couple a distal end of a delivery cable **108** to a medical device **110** (which may be any of the occluders described herein) for facilitating the deployment of medical device **110** at a target site. Medical device **110** is deployed to treat the target site, and, in the example embodiment, is an occlusion device (“occluder”).

[0042] Due to the wide variety of LAA anatomy and possible implant locations for any given implant size, it may be useful to include additional or alternatively-configured stabilizing members (e.g. stabilizing wires) or engagement features (e.g. hooks or hooked ends of a wire) beyond those shown in the embodiments of FIGS. 1-2C to help ensure stability of the occluder in most configurations of anatomy and implant size.

[0043] FIG. 4 is a highly schematic illustration of the medical device **50** used as an occluder that has been deployed within a LAA of a patient. Nearby anatomy, including exemplary locations of the pulmonary vein PV, circumflex artery CX, and mitral valve MV are shown for additional context. In the example configuration of FIG. 4, the occluder **50** has been implanted in a manner that the top of the occluder **50** (in the orientation of the view of FIG. 4) is in contact with most of the tissue on the top side of the device, as indicated at location **1**. However, on the bottom side of the occluder (in the orientation of the view of FIG. 4), only the proximal portion of the lobe **58** is engaged with the tissue. This configuration may result in a scenario in which the engagement features (e.g. hooks) of stabilizing members (e.g. stabilizing wires) **68** of the occluder **50** are not engaged with tissue in that portion of the anatomy, as indicated at location **2**.

[0044] If the type of implantation shown in FIG. 4 occurs, the occluder **50** may be unstable following implantation or deployment, particularly on the bottom side of the occluder **50** (in the orientation of the view of FIG. 4), for example at or near location **2**. During implantation of the occluder **50**, after the occluder **50** has expanded into the LAA but prior to the delivery device **102** being detached from the occluder **50** (e.g. prior to uncoupling the coupling member **106** from the occluder **50**), a traction test or tension test (which may also be referred to as a “tug test”) may be performed. Generally, a traction test (or tension test or tug test) includes pulling proximally on the delivery device **102** to place a slight proximal tension force on the occluder **50** (e.g. via the delivery cable **108**). This slight proximal tension force or “tug” may cause the stabilizing members **68** (and/or engagement features or hooks) to gain better engagement with the surrounding tissue and confirm that the occluder **50** is securely positioned within the LAA (e.g. by confirming that the tension or “tug” does not cause a change in position or orientation of the occluder **50** relative to the LAA despite the applied force). However, if the occluder **50** has the configuration shown in FIG. 4 (or a similar configuration), when the tension or tug test is performed, or if the occluder **50** is left in place under compression and pressure, the occluder can tilt and migrate until the non-engaged stabilizing wires **68** (and/or non-engaged engagement features or hooks) (e.g. at or around location **2**) engage tissue. This scenario can lead to gaps around the sealing surface of the occluder **50**, or even embolization of the occluder **50**.

[0045] FIG. 5 shows an example of an occluder **200** according to an aspect of the disclosure, the occluder **200** being in an expanded condition. In some examples, the occluder **200** is formed of braided mesh (e.g. strands of shape-memory material braided together) or laser cut from a tube of shape-memory material, and may include a proximal disk **210**, a distal lobe **230**, and a connecting

portion **220** connecting the two. Distal lobe **230**, as well as all other distal lobes described herein, may have a generally cylindrical shape or configuration, although one or both ends of the distal lobe **230** may be curved (to varying degrees) and the proximal and distal faces of the distal lobe may not have flat (or at least perfectly flat) shapes. Proximal disk **210**, as well as all other proximal disks described herein, may have a generally flat disk shape or configuration. The distal lobe **230** may be intended for placement within the LAA while the proximal disk **210** may be intended to cover the ostium of the LAA. Compared to occluder **50**, which has stabilizing members **68** with engagement features (e.g. hooks) positioned so that all of the hooks are positioned near the distal face **62**, occluder **200** may include at least a first group of stabilizing members (which may be formed from wire(s)) **240** and a second group of stabilizing members (which may be formed from wire(s)) **250** having a different configuration than the first group. For example, stabilizing members in the first group **240** may be generally similar in size and position as stabilizing members **68**, with engagement features (e.g. hooks or hooked ends of a wire), in particular the distal ends of the engagement features, being positioned near the distal face of the distal lobe **230**. However, stabilizing members in the second group **250** may be smaller and/or shorter, so that the engagement features (e.g. hooks or hooked ends of a wire), in particular the distal ends of the engagement features, are positioned near a mid-length segment (in the axial direction) of the distal lobe **230**. In other words, the engagement features (e.g. hooks) of stabilizing members **250** may be positioned significantly proximal relative to the engagement features (e.g. hooks) of stabilizing members **240**. It should be understood that, in FIG. 5, one stabilizing member **250** is labeled and one stabilizing member **240** is labeled, but a plurality of each type of stabilizing member **240**, **250** may be provided, for example in an alternating configuration around the circumference of the distal lobe **230**. Compared to the stabilizing members **68** of occluder **50**, the configuration of the groups of stabilizing members **240**, **250** of occluder **200** may increase the resilience of the occluder to embolization or motion in an implanted condition similar to that shown in FIG. 4, at least partially due to the second group of stabilizing members **250** being added in a second row, more proximal on the lobe **230** compared to the first group of stabilizing members **240**.

[0046] The location of the engagement features (e.g. hooks or distal hooked ends of wire(s)) of the stabilizing members in the second group **250** may be selected to account for different countervailing objectives. For example, if the engagement features of the stabilizing members in the second group **250** are positioned too far distally, the problem produced in an implanted condition similar to that shown in FIG. 4 may not be solved. However, if the engagement features of the stabilizing members in the second group **250** are positioned too far proximally, they may pose risk to surrounding anatomy such as the circumflex artery CX shown in FIG. 4.

[0047] For devices similar to occluder **50** (and/or occluder **200**), instructions for use may typically indicate that the occluder **50** (or occluder **200**) should be implanted so that at least two-thirds of the length of the lobe **58** (or lobe **230**) is positioned beyond (e.g., distally to) the circumflex artery CX. In such examples, the proximal third of the lobe **58** (or lobe **230**) has the potential to overlap with the circumflex artery CX. To avoid interaction with the circumflex artery CX, in one design, engagement features (e.g. hooks or hooked ends of wire(s)) of the stabilizing members in the second group **250** may be positioned a minimum of the proximal third of the length of the lobe **58** (or lobe **230**). In other words, if the axial length of the lobe **58** (or lobe **230**) from the proximal edge to the distal edge is a unit of length, the engagement features of the stabilizing members of the second group **250** may be positioned a minimum of one-third of that unit of length beyond the proximal edge of the lobe **58** (or lobe **230**) (that is, the stabilizing members in the second group **250** may be positioned at least 33% of the axial length beyond the proximal face of the distal lobe **230**) to prevent potential damage to the circumflex artery CX if the engagement features partially penetrate into the tissue wall of the heart. In some examples, the lengths described in this paragraph refer to dimensions when the occluder **50** (or occluder **200**) is in an unbiased condition, for example in the absence of applied forces. Further, when referring to the stabilizing members **250**

being positioned at least 33% of the axial length beyond the proximal face of the distal lobe **230**, the structure of the stabilizing member **250** having such positioning is preferably the tip (e.g. the terminal end configured to actually frictionally engage tissue), although in other embodiments the structure of the stabilizing member **250** having such positioning may be the distal-most end of the stabilizing member **250**.

[0048] For example, FIG. **6** is another view of occluder **200** similar to FIG. **5**, showing a dimension A extending from the proximal face of the lobe **230** toward the distal face. The dimension A may represent the distance between the proximal face of the lobe **230** and the position of the tips or terminal ends of the engagement features (e.g. hooks or hooked ends of wire(s)) of the stabilizing members in the second group **250**. As noted above, the dimension A preferably has a length of at least one-third the total length of the lobe **230**, the total length being represented by the dimension B in FIG. **6**. In some examples, occluders **200** may be provided in various sizes, which may include different sized lobes. For example, a small size occluder **200** may include a lobe having a length B of about 7.5 mm, while a large size occluder **200** may include a lobe having a length B of about 10 mm. In some examples, for the small size occluder **200**, the dimension A may be about $3.5\text{ mm}\pm 0.5\text{ mm}$, while for the large size occluder **200**, the dimension A may be about $4.5\text{ mm}\pm 0.75\text{ mm}$. In some examples, the dimension A may be between about 33% and about 55% of the dimension B, including for example about 35%, about 40%, about 45%, and about 50%. In other configurations, including for example those shown in FIGS. **14-17** and described in greater detail below, the proximal dimension A can vary slightly per pair of stabilizing members, which may create a lower required force when recapturing the device because the engagement features (e.g. hooks or hooked ends of wire(s)) do not all contact the delivery sheath at the same time, as they are offset at least slightly.

[0049] It should be understood that the stabilizing members and/or the engagement features (e.g. hooks) thereof described herein may take any suitable form. Two examples of a stabilizing members **300** are shown in FIGS. **7A-B**. Additional detail regarding the specific structure of stabilizing member **300**, and other suitable options for stabilizing members, is described in greater detail in U.S. Patent Application Publication No. 2022/0280166, the disclosure of which is hereby incorporated by reference herein. In the examples of FIGS. **7A-B**, stabilizing member **300** may be formed from wire or other material described herein, or formed from a laser cut design as described herein, and may include at least one eyelet **307**. The stabilizing member **300** may generally include a backing portion **304**, which may include a general “U”-shape design positioned opposite the engagement features, with two legs extending from the backing portion **304** to the opposite engagement features **302**, which may be a hook, a hooked end, or tip configured to engage tissue. The eyelet(s) **307**, if included, may be used to couple (e.g., via suturing) the stabilizing member **300** to the occluder (e.g., occluder **50** or **200**). As should be understood, features like the length of the stabilizing member **300** between the backing portion **304** and the engagement features **302** may be adjusted as desired to achieve the desired effect (e.g., a shorter length for use in the group of stabilizing members **250** of FIGS. **5-6**, or a longer length for use in the group of stabilizing members **240** of FIGS. **5-6**). In some examples, particularly for longer lobes, the different groups of stabilizing members may have equal lengths.

[0050] One factor for consideration described above is the axial position of the engagement features of the one or more groups of stabilizing members (e.g., groups **240**, **250**) relative to the length of lobe **230** and relative to the eventual implanted position within the LAA. Another factor in the design is the radial spacing of the stabilizing members. For example, in the embodiment shown in FIGS. **5-6**, each stabilizing member in the second group **250** is positioned between a pair of stabilizing members in the first group **240**, with the circumferential spacing between each adjacent stabilizing member being substantially even. In other words, in the embodiment of FIGS. **5-6**, the stabilizing members are positioned to create substantially even circumferential spacing between the stabilizing members so that occluder **200** has no or low sensitivity to rotational

orientation within the LAA. Typical delivery systems for LAA occlusion devices do not include the capability to control the rotation of the occluder (the rotation being defined about a longitudinal axis passing through the center of the connection portion **220** in a proximal-to-distal direction, for example). Additionally, there is large variety of LAA shapes among patients, and many LAA anatomies are elliptical or non-symmetric. Thus, it may actually be preferable to not require any rotational control of the occluder **200**, and as such, it may be important for the occluder **200** to be rotationally symmetric about one or more planes to reduce or eliminate the impact of implantation in one rotational position compared to another rotational position. In some examples, the angle between each adjacent pair of stabilizing members in the first group **240**, and/or between each adjacent pair of stabilizing members in the second group **250**, is between about 20 degrees and about 65 degrees. However, it should be understood that for larger size versions of the occluder **200** (which may have more total stabilizing members), the angle may be smaller, and for smaller size versions of the occluder **200** (which may have fewer total stabilizing members), the angle may be larger.

[0051] FIG. **8** is an enlarged view of a stabilizing member **400** (e.g. stabilizing wire) coupled to a lobe of an occluder. In this particular example, stabilizing member **400** has the general design shown in FIGS. **7A-B** and is a smaller stabilizing member of the group of stabilizing members **250** of FIG. **5**. In the view of FIG. **8**, the top of the figure is the proximal face of the lobe **230**, with the distal face of the lobe not being shown but being positioned in the direction of the bottom of the figure. As can be seen, in this example, the backing portion **404** of the stabilizing member **400** is positioned radially outward of the braided mesh that forms the lobe **230**, with the two legs extending from the backing portion **404** being largely positioned radially inside the lobe **230**. A connector **410**, which in this case is a suture or other string-like member, connects, tics, or otherwise couples each eyelet **407** to a portion of the braid that forms the lobe **230**. The engagement features **402** (e.g. hooks, hooked ends, or tips) of the stabilizing members **400** extend through an opening in the braided mesh forming the lobe **230** so that the engagement features **402** extend from the radially inner portion of the lobe **230** radially outward of the lobe **230** so that the terminal ends are available for engagement with tissue upon implantation. In some examples, although stabilizing member **400** represents the shorter stabilizing member of the second group **250** of FIGS. **5-6**, the stabilizing members used for the first group **240** of FIGS. **5-6** may be similar or identical from the eyelets **407** to the engagement features **402**, with the only significant difference being the length of the legs between the backing portion **404** and the eyelets **407**.

[0052] FIG. **9A** is another view of an occluder **200** which may be similar or identical to that shown in FIGS. **5A-B**, which may include a proximal disk **210** connected to a distal lobe **230** by a connecting portion **220**. In this particular example, the smaller stabilizing members **250** within the second group may be formed by laser cutting the stabilizing members **250** from a sheet of material (e.g., nitinol) and then setting the stabilizing member **250** into the desired shape (e.g., similar to that shown in FIG. **7A**). FIG. **9B** shows an example of the cut pattern for the smaller stabilizing member **250** before being shaped or contoured. For example, the stabilizing member of FIG. **9B** can be laser cut and then shaped to the desired shape, or in other examples formed from a wire and then shaped to the desired shape. The view of FIG. **9B** illustrates examples of positions of the backing portion **254**, two eyelets **257**, and two engagement features (e.g. hooks or hooked ends) **252**, prior to being shaped (or shape-set) into the corresponding features shown in, for example, FIG. **7A**.

[0053] FIG. **10A** is another example of an occluder with two rows of engagement features of stabilizing members at different axial positions. For example, instead of having two groups of different stabilizing members with engagement features (e.g. hooks or hooked ends) at different positions, the example of FIG. **10A** includes stabilizing members that are all similar or identical to each other, but that each include one or more engagement features closer to the distal end of the lobe, as well as one or more additional engagement features nearer the mid-length segment of the

distal lobe. In this example, occluder **500** may be substantially similar or identical to occluder **200** with the exception of the stabilizing members. For example, occluder **500** may include a proximal disk **510** connected to a distal lobe **530** by a connection portion **520**. In some examples, each stabilizing member **550** may include a backing portion **554** and two distal engagement features (e.g. distal hooks) **552** configured to engage tissue. A pair of proximal engagement features (e.g. proximal hooks) **556** may be positioned on the legs extending between the backing portion **554** and the distal engagement features **552**. As should be understood, the relationship between the position of the distal engagement features **552** and the proximal engagement features **556** may be similar or identical to the engagement features of the stabilizing members **240** in the first group of occluder **200** and the engagement features of the stabilizing members **250** in the second group of occluder **200**.

[0054] FIGS. **10B-10D** show examples of patterns for the stabilizing member before being shaped or contoured, which show examples of how the different engagement features (e.g. hooked ends) may be positioned relative to each other before being contoured. For example, FIG. **10B** shows one example of a cut pattern **550a** that may be used, for example by laser-cutting a flat sheet of material (e.g., nitinol), to create a structure that may then be used to form stabilizing member **550**, for example by shape-setting the cut structure. As shown in FIG. **10B**, the cut pattern **550a** may include a portion **554a** that may be contoured into a backing portion, a pair of proximal engagement features (e.g. proximal hooks) **556a**, a pair of distal engagement features (e.g. distal hooks) **552a**, and a pair of eyelets **557a** each positioned between one of the proximal engagement features (e.g. proximal hooks) **556a** and one of the distal engagement features (e.g. distal hooks) **552a** on the same leg of the stabilizing member.

[0055] FIG. **10C** shows another example of a cut pattern **550b** that may be used, for example by laser-cutting a flat sheet of material (e.g., nitinol), to create a structure that may then be used to form stabilizing member **550**, for example by shape-setting the cut structure. As shown in FIG. **10C**, the cut pattern **550b** may include a portion **554b** that may be contoured into a backing portion, a pair of proximal engagement features (e.g. proximal hooks) **556b**, a pair of distal engagement features (e.g. distal hooks **552b**), and a pair of eyelets **557b** each positioned between one of the proximal engagement features **556b** and one of the distal engagement features **552b** on the same leg of the stabilizing member. Cut pattern **550b** is generally similar to cut pattern **550a**, with two main differences. First, while each eyelet **557a** of cut pattern **550a** is formed entirely on one side of the leg extending between the backing portion **554a** and the distal engagement feature **552a**, each eyelet **557b** of cut pattern **550b** provides a transition of the leg extending between the backing portion **554b** and the distal engagement feature **552b**. Stated otherwise, each leg extending from the backing portion **554b** meets an eyelet **557b** on a rear portion of the eyelet **557b**, with the leg continuing to extend from the front portion of the eyelet **557b** to form the distal engagement feature **552b**. With this configuration, the distal engagement features **552b** may be positioned generally aligned with the proximal engagement features **556b**, despite the second main difference, described below. Cut pattern **550b** may include a more complete proximal engagement feature (e.g. proximal hook) **556b** compared to proximal engagement feature (e.g. proximal hook) **556a**. For example, whereas the hooked portion of proximal engagement feature **556a** extends directly from the leg extending between the backing portion **554a** and the distal engagement feature **552a**, the hooked portion of proximal engagement feature **556b** may include an extension so that the shape of the proximal engagement feature **556b** substantially matches the shape of distal engagement feature **552b**. This may provide increased uniformity between the proximal engagement features **556b** and the distal engagement features **552b**.

[0056] FIG. **10D** shows another example of a cut pattern **550c** that may be used, for example by laser-cutting a flat sheet of material (e.g., nitinol), to create a structure that may then be used to form stabilizing member **550**, for example by shape-setting the cut structure. Cut pattern **550c** is identical to cut pattern **550b** in most respects, including in that it includes a backing portion **554c**,

proximal engagement features (e.g. proximal hooks) **556c** and distal engagement features (e.g. distal hooks) **552c**, and eyelets **557c**. While each of these individual components of cut pattern **550c** may have substantially identical shapes to their counterparts in cut pattern **550b**, the main difference between cut pattern **550c** and cut pattern **550b** is the position of the proximal engagement feature **556c** relative to the adjacent portion of the leg extending between the backing portion **554c** and eyelet **557c**. For example, in cut pattern **550c**, the proximal engagement feature **556c** has more spacing from the adjacent leg compared to the spacing between proximal engagement feature **556b** and its adjacent leg. This may be achieved, in one example, by including a more acute angle where the leg emerges from the backing portion **554c**, compared to the smoother transition between backing portion **554b** and the adjacent leg in cut pattern **550b**.

[0057] Regardless of whether cut pattern **550a**, **550b**, **550c**, or another cut pattern is used to form stabilizing members **550**, the resulting configuration may help ensure that there are an equal number of distal engagement features (e.g. distal hooks) **552** as proximal engagement features (e.g. proximal hooks) **556**, and also help ensure consistent spacing between the distal engagement features **552** and proximal engagement features **556**. In embodiments that use cut pattern **550b** or **550c**, or similar cut patterns, the additional clearance provided adjacent the proximal engagement features **556b**, **556c** may facilitate relatively independent unfurling of the proximal engagement features **556** during recapture of the occluder **500** within a delivery device. The use of stabilizing members **550** may also reduce complexity, at least in that separate types of stabilizing members (e.g. one group of large stabilizing members and one group of small stabilizing members) may be avoided.

[0058] FIG. **11** is another example of an occluder with two rows of engagement features of stabilizing members at different axial positions. In this example, each stabilizing member has a pair of engagement features (e.g. hooks or hooked ends), but one of the two engagement features has a relatively distal position while the other of the two engagement features has a relatively proximal position. In this example, occluder **600** may be substantially similar or identical to occluder **200** with the exception of the stabilizing members. For example, occluder **600** may include a proximal disk **610** connected to a distal lobe **630** by a connection portion **620**. In some examples, each stabilizing member **650** may include a backing portion **654**, one distal engagement feature (e.g. distal hook) **652** configured to engage tissue, and one proximal engagement feature (e.g. proximal hook) **656**. As should be understood, the relationship between the position of the distal engagement feature **652** and the proximal engagement feature **656** may be similar or identical to the engagement features of the stabilizing members **240** in the first group of occluder **200** and the engagement features of the stabilizing members **250** in the second group of occluder **200**. Whereas other stabilizing members described above are generally symmetric about a centerline, stabilizing members **650** are not. For example, stabilizing members **650** may have an offset configuration in which one relatively short leg extends from backing portion **654** to proximal engagement features **656**, and one relatively long leg extends from backing portion **654** to distal engagement feature **652**. In this example, the use of stabilizing members **650** may help ensure that there are an equal number of distal engagement features **652** as proximal engagement features **656**, and also help ensure consistent spacing between the distal engagement features **652** and proximal engagement features **656**. Further, the use of stabilizing members **650** may also reduce complexity, at least in that separate types of stabilizing members (e.g. one group of large stabilizing members and one group of small stabilizing members) may be avoided. Although not shown, stabilizing members **650** may include one or more eyelets to assist with connecting (e.g., by sutures) the stabilizing members **650** to the occluder **600**.

[0059] FIG. **12A** is another example of an occluder with two rows of engagement features of stabilizing members at different axial positions. In this embodiment, the stabilizing members may each have a pair of engagement features (e.g. hooks or hooked ends) positioned relatively distally, with an individual engagement feature (e.g. hook or hooked end) extending from the center portion

of the backing portion so that individual engagement features is positioned proximal to the pair of engagement features. In this example, occluder **700** may be substantially similar or identical to occluder **200** with the exception of the stabilizing members. For example, occluder **700** may include a proximal disk **710** connected to a distal lobe **730** by a connection portion **720**. In some examples, each stabilizing member **750** may include a backing portion **754**, two distal engagement features (e.g. distal hooks) **752** configured to engage tissue, and a single proximal engagement feature (e.g. proximal hook) **756** extending from the backing portion **754** and generally centered between the legs that extend to the distal engagement features **752**. As should be understood, the relationship between the position of the distal engagement features **752** and the proximal engagement feature **756** may be similar or identical to the engagement features of the stabilizing members **240** in the first group of occluder **200** and the engagement features of the stabilizing members **250** in the second group of occluder **200**.

[0060] FIGS. **12B-12C** show examples of the stabilizing members that may achieve this three-engagement-feature embodiment. For example, FIG. **12B** shows one example of a stabilizing member **750a** that may be used as one or more of the stabilizing members **750**. As shown in FIG. **12B**, the stabilizing member **750a** include a backing portion **754a** that may include a connector **755a**. In the illustrated example, connector **755a** may take the form of two prongs extending transverse to the backing portion **754a**, with the two prongs extending substantially parallel to each other with a space therebetween to define a recess, although in other embodiments the connector **755a** may take other forms. Two legs may each extend from the backing portion **754a** to an eyelet **757a** and then to a distal engagement feature (e.g. distal hook) **752a**, for example similar to as shown in FIGS. **7A-7B**. Stabilizing member **750a** may also include a separately formed proximal engagement feature (e.g. proximal hook **756a**) that is coupled to the backing portion **754a**. For example, the proximal engagement feature **756a** may include its own eyelet **758a** and its own connector **759a**, the eyelet **758a** being positioned between the connector **759a** and the tip of the proximal engagement feature **756a**. In the illustrated example, the connector **759a** may have a generally complementary shape to connector **755a**. For example, the connector **759a** may take the form of two prongs extending away from the eyelet **758a**, with the two prongs extending substantially parallel to each other with a space therebetween to define a recess, although in other embodiments the connector **759a** may take other forms. The pair of prongs of one or both of connectors **755a**, **759a** may be angled slightly toward each other near the free ends thereof. The two connectors **755a**, **759a** may be coupled to each other so that a portion of the backing member **754a** is received within the recess between the prongs of connector **759a**, with a portion of connector **759a** received within the recess between the prongs of connector **755a**. If one or both pairs of prongs are provided with a taper or angle, a clamping force may result upon coupling the connectors **755a**, **759a** to help the construct remain connected, although alternatively or in addition, sutures or other members may be used (e.g. including using eyelet **758a**) to help the connectors **755a**, **759a** to remain coupled to each other.

[0061] Whereas the stabilizing member **750a** of FIG. **12B** includes an interlocking proximal engagement feature (e.g. proximal hook) **756a**, in other embodiments, stabilizing member **750** may be formed as a single unitary member, such as a single laser-cut stabilizing member **750b** as shown in FIG. **12C**. Stabilizing member **750b** includes the same main component as stabilizing member **750a**, such as two legs each extending from a backing portion **754b** to an eyelet **757b** and then to a distal engagement feature (e.g. distal hook) **752b**, for example similar to as shown in FIGS. **7A-7B**. Stabilizing member **750b** may also include a integrally formed proximal engagement feature (e.g. proximal hook) **756b** that extends from the backing portion **754b**. In some examples, the proximal engagement feature **756b** may include its own eyelet **758b**. In other words, stabilizing member **750b** may be substantially identical to stabilizing member **750a**, except that stabilizing member **750b** omits the connectors because stabilizing member **750b** may be formed as a single integral member.

[0062] Whether stabilizing members **750** take the form shown in FIG. **12B**, FIG. **12C**, or another form, the resulting configuration may be such that the proximal row of engagement features (e.g. proximal hooks) has one half of the total number of distal engagement features (e.g. distal hooks), unlike other embodiments described herein. Also, unlike certain other embodiments described herein, the inclusion of both a proximal engagement feature and distal engagement features on a single assembled or formed stabilizing member may result in some amount of dependency between the hooks. For example, anchoring of the proximal (or distal) engagement features into tissue may affect the ability of the other engagement features on the same assembled or formed stabilizing member to anchor into tissue. Further, as with certain other embodiments described herein, fewer total stabilizing members may be required when providing both proximal and distal engagement features on the same stabilizing member, which may reduce complexity compared to embodiments that require proximal and distal engagement features to be positioned on separate stabilizing members. It should be understood that, in some embodiments, the position of the proximal engagement members relative to the distal engagement members in the axial direction and relative to the length of the lobe **730** is substantially the same as in other embodiments, such as that shown and described in connection with FIG. **6**. However, in some embodiments, the gaps between proximal engagement features **756** and distal engagement features **752** may be even larger than in other embodiments. For example, in one embodiment in which the axial length of the lobe **730** is about 7.5 mm, the proximal engagement features **756** are positioned about 2.5 mm distally (e.g. about $\frac{1}{3}$ or 33%) of the proximal face of the lobe **730**.

[0063] FIG. **13** is another example of an occluder with two rows of engagement features of stabilizing members at different axial positions. This example is similar to that shown in FIGS. **5-6**, but in this case all of the stabilizing members have the smaller configuration (e.g., similar to the group of stabilizing members **250**), with the stabilizing members positioned in two rows—one relatively distal and one relatively proximal. In this example, occluder **800** may be substantially similar or identical to occluder **200** with the exception of at least some of the stabilizing members. For example, occluder **800** may include a proximal disk **810** connected to a distal lobe **830** by a connection portion **820**. Occluder **800** may include a first group of stabilizing members **840** and a second group of stabilizing members **850**. Generally, the first group of stabilizing members **840**, which include relatively distally positioned engagement features (e.g. hooks), may correspond to the first group of stabilizing members **240** of occluder **200**, whereas the second group of stabilizing members **850**, which include relatively proximally positioned engagement features (e.g. hooks), may correspond to the second group of stabilizing members **250** of occluder **200**. In some examples, stabilizing members **850** may be similar or identical to stabilizing members **250**, including being formed as a relatively small stabilizing member, such as the example shown and described in connection with FIG. **7A**. However, unlike the relatively long stabilizing members **240** of occluder **200**, the stabilizing members **840** of occluder **800** may also be relatively small stabilizing members. For example, stabilizing members **840** may be similar or identical in structure to stabilizing members **850**, and only different in their relative positions on the lobe **830**. The positioning of the proximal stabilizing members **850** relative to the length of the lobe **830** and relative to the distal stabilizing members **840** may be substantially similar or identical as described in connection with occluder **200**.

[0064] Although many of the embodiments described herein include stabilizing members that include two groups of engagements features (e.g. hooks or hooked ends) (including individual stabilizing members with two axially offset engagement features, or separate stabilizing members being positioned so that engagement features are axially offset), it should be understood that additional stabilizing members and/or engagement features may be provided beyond the two groups. For example, any of the embodiments described herein may include additional stabilizing members and/or additional engagement features on the stabilizing members already provided so that there are three or more (e.g. four, five, six or more) groups of engagement features provided at

different axial locations on the occluder.

[0065] In some examples, an occluder may be provided with stabilizing members (and/or engagement features) with a pattern that is dependent, at least in part, on the size of the device. For example, for an occluder that includes a lobe and a disk that is intended for implantation into the LAA, the occluder may be offered in eight different sizes (although more or fewer sizes may be suitable). In one example, an occluder may be offered in two size groups, including a “small” and “large” size group. For example, a “small” size group may include a lobe that has a length of about 7.5 mm and a “large” size group may include a lobe that has a length of about 10 mm. In some examples, the “small” size group may include four sizes with respective lobe and disk diameters of (1) about 16 mm and about 22 mm, (2) about 18 mm and about 24 mm, (3) about 20 mm and about 26 mm, and (4) about 22 mm and about 28 mm. In some examples, the “large” size group may include four sizes with respective lobe and disk diameters of (5) about 25 mm and about 32 mm, (6) about 28 mm and about 35 mm, (7) about 31 mm and about 38 mm, and (8) about 34 mm and about 41 mm. The different sizes of the occluder, including for example the different lengths of the lobes of the occluders, may make it desirable to have a different stabilizing member (and/or engagement feature) configuration depending on the length of the lobe. It should be understood that all dimensions provided in this paragraph are based on the occluder in the absence of applied forces (e.g. in the natural or set-shape).

[0066] As an example of the above point that different size occluder devices may be provided with different configurations of stabilizing members (and/or engagement features), a “small” size LAA occluder, with “small” being defined as a relatively small length (e.g. about 7.5 mm) lobe, may be provided with two groups of stabilizing members similar to that shown in FIG. 6. In this type of example, the occluder **200** may include a first row or group of stabilizing members **240** that are relatively large compared to a second row or group of stabilizing members **250**. In this example, the engagement features (e.g. hooks or hooked ends), or the terminal ends or tips thereof, of the smaller stabilizing members **250** in the second group may be positioned between about 3.3 mm to about 4.4 mm, including about 3.8 mm, distal of the proximal plane or surface of the lobe **230** (e.g. the distance labeled “A” in FIG. 6). In this example, the engagement features (e.g. hooks or hooked ends), or the terminal ends or tips thereof, of the larger stabilizing members **240** in the first group may be positioned between about 0 mm and about 2 mm, including about 1 mm, proximal of the distal plane of surface of the lobe **230**. However, for a “large” size LAA occluder that has a larger length lobe (e.g. about 10 mm compared to about 7.5 mm), it may be useful to provide stabilizing members in additional and/or different positions relative to the “small” size LAA occluder, which may be useful for providing anchoring or stabilization at additional locations. In other words, because “large” occluders are intended for use in patients with larger LAA anatomies, additional securement may be desirable (compared to “small” occluders that are intended for use in patients with smaller LAA anatomies) to account for the increased amount of tissue that the “large” occluder may need to contact and/or traverse. Providing staggered positioning of the stabilizing members, and particularly staggering the axial position of the engagement features (e.g. hooks or hooked ends) of the stabilizing members, may provide another benefit independent of that which is described immediately above. In some examples of use, the occluder (e.g. occluder **200**) may need to be pulled back into the delivery device (e.g. delivery catheter **104**) for repositioning after the occluder has been partially deployed (e.g. while the lobe **230** has been deployed but while the disk **210** is still coupled to the delivery device, such as to delivery cable **108**). If resheathing is performed after the engagement features (e.g. hooks or hooked ends) of the stabilizing members have already deployed from the delivery device, the engagement features of the stabilizing wires may “catch” or “grab” onto the distal end of the delivery catheter **104**, increasing the amount of force needed to fully resheath the occluder **200** into the delivery catheter **104**. However, if the axial position of these engagement members is staggered, fewer engagement members will simultaneously “catch” or “grab” onto the distal end of the delivery catheter **104** during resheathing

at any one particular time, which may reduce the force and/or difficulty required to fully resheath the occluder **200**.

[0067] One example of an occluder **900** for use in the LAA is shown in FIGS. **14-17**. Occluder **900** may be substantially similar or identical to other occluders described herein, including for example occluder **200**, with exceptions regarding the positions of the stabilizing members (and/or engagement features) and/or the length of the lobe. FIGS. **14-15** illustrate top perspective, and bottom views of occluder **900**, with the braided mesh forming the disk **910** and the lobe **930** depicted as a solid surface to better illustrate the positions of the various stabilizing members (and/or engagement features). FIG. **16** is a more detailed side view of the occluder **900**, while FIG. **17** provides a simplified view of the occluder **900**.

[0068] Referring to FIGS. **14-17**, occluder **900** may be a “large” size occluder as described above, and include a proximal disk **910** connected to a distal lobe **930**, for example via a connecting portion **920** (which is visible in the view of FIG. **16-17**). As noted above, both disk **910** and lobe **930** may be formed of a braided mesh (e.g. strands of nitinol braided together), but the disk **910** and lobe **930** are shown as solid in FIGS. **14-15** for clarity of illustration. In this example, three separate groups of stabilizing members are coupled to the lobe **930**. In the particular illustrated example, lobe **930** includes a first group of stabilizing members **940** which may be similar or identical to the first group of stabilizing members **240**. For example, each stabilizing member **940** in the first group may have the same basic shape or configuration as shown in FIG. **7A** or FIG. **7B**, with a proximal backing portion of the stabilizing member **940** being positioned radially outward of the lobe **930**, two legs of the stabilizing member **940** extending distally radially inwardly of the lobe **930**, and two engagement features (e.g. hooks or hooked ends) passing through the lobe **930** to be positioned radially outward of the lobe **930** and hooking back in the proximal direction toward the disk **910**. In some examples, the tips of the stabilizing members **940** (e.g. tips of the engagement features, which may be hooks) are positioned between about 0 mm and about 2 mm, including about 1 mm, proximal to the distal face of the lobe **930**. In the illustrated embodiment, every other stabilizing member on the lobe **930** is a part of the first group of stabilizing members **940**.

[0069] In the particular illustrated example of occluder **900**, lobe **930** includes a second group of stabilizing members **950** which may be similar or identical to the second group of stabilizing members **250**. For example, each stabilizing member **950** in the second group may have the same basic shape or configuration as shown in FIG. **7A** or FIG. **7B**, with a proximal backing portion of the stabilizing member **950** being positioned radially outward of the lobe **930**, two legs of the stabilizing member **950** extending distally radially inwardly of the lobe **930**, and two engagement features (e.g. hooks or hooked ends) passing through the lobe **930** to be positioned radially outward of the lobe **930** and hooking back in the proximal direction toward the disk **910**. In some examples, the tips of the stabilizing members **950** (e.g. the tips of the engagement features, which may be hooks) are positioned between about 3.3 mm and about 4.3 mm, including about 3.8 mm, distal to the proximal face of the lobe **930**. In the illustrated embodiment, every fourth stabilizing member on the lobe **930** is a part of the second group of stabilizing members **950**.

[0070] In the particular illustrated example of occluder **900**, lobe **930** includes a third group of stabilizing members **960** which may be similar or identical to the second group of stabilizing members **250**, albeit with different positioning as described below. For example, each stabilizing member **960** in the third group may have the same basic shape or configuration as shown in FIG. **7A** or FIG. **7B**, with a proximal backing portion of the stabilizing member **960** being positioned radially outward of the lobe **930**, two legs of the stabilizing member **960** extending distally radially inwardly of the lobe **930**, and two engagement features (e.g. hooks or hooked ends) passing through the lobe **930** to be positioned radially outward of the lobe **930** and hooking back in the proximal direction toward the disk **910**. In some examples, the tips of the stabilizing members **960** (e.g. the tips of the engagement features, which may be hooks) are positioned between about 4.4

mm and about 5.4 mm, including about 4.9 mm, distal to the proximal face of the lobe **930**. In the illustrated embodiment, every fourth stabilizing member on the lobe **930** is a part of the third group of stabilizing members **960**.

[0071] As best shown in FIGS. **16-17**, with the particular configuration described above, the lobe **930** includes larger stabilizing members **940** alternating with smaller stabilizing members around the circumference of the lobe **930**, with the smaller stabilizing members themselves alternating between a stabilizing member **950** from the second group alternating with a stabilizing member **960** from the third group, such that the order of stabilizing members around the circumference of the lobe is stabilizing member **940**, stabilizing member **950**, stabilizing member **940**, stabilizing member **960**, stabilizing member **940**, stabilizing member **950** and so forth. With this configuration, the positions of the tips of the stabilizing members **940**, **950**, **960** (e.g. the tips of the engagement features, which may be hooks) are more varied along the length of the lobe **930** to increase the likelihood of getting good frictional engagement between the various stabilizing members **940**, **950**, **960** and the tissue of the LAA. This may be desirable given the large length of the lobe **930** that incorporates the three different groups of stabilizing members **940**, **950**, **960**. Further, as noted above, during a resheathing operation, the engagement features (e.g. hooks) of stabilizing members **950** may be the first group to contact the distal end of the delivery catheter **104**, and after that the engagement features (e.g. hooks) of stabilizing members **960** are the second group to contact the distal end of the delivery catheter **104**, and finally the engagement features (e.g. hooks) of stabilizing members **940** are the third and final group to contact the distal end of the delivery catheter **104**. This may reduce the force the user needs to create to complete recapture compared to an embodiment in which the engagement features of stabilizing members **950** and **960** all contact the distal end of the delivery catheter **104** simultaneously during resheathing. For occluders with lobes having smaller lengths, the third group of stabilizing members **960** may be replaced with the second group of stabilizing members **950**, resulting in the configuration shown in, and described in connection with, FIGS. **5-6**. It should also be understood that, in the embodiment of FIGS. **14-17**, the proximal ends of each group of stabilizing members **940**, **950**, **960** are not positioned at the same axial distance as any other of the groups of stabilizing members, which may help to create less packing in these regions during compression of the occluder **900** within a delivery device.

[0072] Although the invention herein has been described with reference to particular embodiments, it is to be understood that these embodiments are merely illustrative of the principles and applications of the present invention. It is therefore to be understood that numerous modifications may be made to the illustrative embodiments and that other arrangements may be devised without departing from the spirit and scope of the present invention as defined by the appended claims.

Claims

1. A medical device for occluding a left atrial appendage (“LAA”), the medical device comprising: a proximal disk configured to cover an ostium of the LAA in an implanted condition of the medical device; a distal lobe configured to be received within a cavity of the LAA in the implanted condition of the medical device; a connecting member connecting the proximal disk to the distal lobe; a first group of stabilizing members, each stabilizing member in the first group having a first engagement feature configured to engage tissue surrounding the LAA in the implanted condition of the medical device; and a second group of stabilizing members, each stabilizing member in the second group having a second engagement feature configured to engage tissue surrounding the LAA in the implanted condition of the medical device, wherein in an unbiased condition of the medical device, the distal lobe has an axial length extending between a proximal face and a distal face of the distal lobe, the second engagement features of the stabilizing members in the second group being positioned at least 33% of the axial length beyond the proximal face of the distal lobe,

the first engagement features of the stabilizing members in the first group being positioned distal to the second engagement features of the stabilizing members in the second group.

2. The medical device of claim 1, wherein the second engagement features of the stabilizing members in the second group are positioned between about 33% and about 55% of the axial length beyond the proximal face of the distal lobe.

3. The medical device of claim 1, wherein the axial length is about 7.5 mm, and the second engagement features of the stabilizing members in the second group are positioned between about 3.0 mm and about 4.0 mm beyond the proximal face of the distal lobe.

4. The medical device of claim 1, wherein the axial length is about 10 mm, and the second engagement features of the stabilizing members in the second group are positioned between about 3.75 mm and about 5.25 mm beyond the proximal face of the distal lobe.

5. The medical device of claim 1, wherein each stabilizing member in the first group is positioned between a pair of stabilizing members in the second group.

6. The medical device of claim 1, wherein each stabilizing member in the first group is substantially identical in structure to each stabilizing member in the second group.

7. The medical device of claim 1, wherein each stabilizing member in the first group has an axial length that is equal to or longer than an axial length of each stabilizing member in the second group.

8. The medical device of claim 1, wherein a total number of the stabilizing members in the first group is equal to a total number of stabilizing members in the second group.

9. The medical device of claim 1, wherein a spacing between each adjacent pair of stabilizing members is substantially equal.

10. The medical device of claim 1, wherein a radial spacing between each adjacent pair of stabilizing members is between about 20 degrees and about 65 degrees.

11. A medical device for occluding a left atrial appendage ("LAA"), the medical device comprising: a proximal disk configured to cover an ostium of the LAA in an implanted condition of the medical device; a distal lobe configured to be received within a cavity of the LAA in the implanted condition of the medical device; a connecting member connecting the proximal disk to the distal lobe; a first group of stabilizing members, each stabilizing member in the first group having a first engagement feature configured to engage tissue surrounding the LAA in the implanted condition of the medical device; and a second group of stabilizing members, each stabilizing member in the second group having a second engagement feature configured to engage tissue surrounding the LAA in the implanted condition of the medical device; and a third group of stabilizing members, each stabilizing member in the third group having a third engagement feature configured to engage tissue surrounding the LAA in the implanted condition of the medical device, wherein in an unbiased condition of the medical device, the distal lobe has an axial length extending between a proximal face and a distal face of the distal lobe, the first engagement features are positioned at a first distance along the axial length, the second engagement features are positioned at a second distance along the axial length, and the third engagement features are positioned at a third distance along the axial length, the first distance being different than the second distance, the second distance being different than the third distance, the third distance being different than the first distance.

12. The medical device of claim 11, wherein the axial length is about 10 mm.

13. The medical device of claim 12, wherein the first distance is between about 8 mm and about 10 mm, such that the first engagement features are positioned between about 0 mm and about 2 mm proximally of the distal face of the distal lobe.

14. The medical device of claim 13, wherein the second distance is between about 3.3 mm and about 4.3 mm.

15. The medical device of claim 14, wherein the third distance is between about 4.4 mm and about 5.4 mm.

- 16.** The medical device of claim 11, wherein the medical device includes a total number of stabilizing members in the first group that is twice as many as a total number of stabilizing members in the second group and that is twice as many as a total number of stabilizing members in the third group.
- 17.** The medical device of claim 11, wherein a single stabilizing member in either the second group or the third group is positioned between each circumferentially adjacent pair of stabilizing members in the first group.
- 18.** The medical device of claim 11, wherein each stabilizing member is separate from the other stabilizing members, wherein each group of stabilizing members comprises more than one stabilizing member, and wherein each stabilizing member is connected to the distal lobe.
- 19.** The medical device of claim 11, wherein each stabilizing member in the first, second, and third groups is either (i) formed of a wire or (ii) cut from a sheet of material.
- 20.** The medical device of claim 11, wherein each first engagement feature, each second engagement feature, and each third engagement feature is a hook.
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